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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

March 2017 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-MAR17

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Insert

Instructions

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/answer book.

4368

QAN 500/8702/2

IB/M/Mar17/E3

Data Sheet for Tenpin bowling



The game

A game has ten **frames** for each player.

In each frame, a player starts with ten pins to knock over.

A player bowls a ball one **or** two times in a frame.

If they knock over all ten pins with their first ball, their frame is complete.
This is called a **strike**.

If there are pins still standing, they bowl a second ball at the remaining pins.
If they knock them all over, this is called a **spare**.

Scoring

These examples show how to work out a player's score.

Example 1 – Sophie's first two frames

Frame 1

1st ball → 6 pins 2nd ball → 2 pins

Frame 2

1st ball → 5 pins 2nd ball → 4 pins

Here is the scoreboard after Frame 2

1		2	
6	2	5	4
8		17	
↑ 6 + 2		↑ 8 + 5 + 4	

After Frame 1, Sophie's total score is 8
After Frame 2, Sophie's total score is 17

Example 2 – Mike's first two frames

Frame 1

1st ball → 3 pins 2nd ball → 7 pins

spare

Frame 2

1st ball → 6 pins 2nd ball → 2 pins

Here is the scoreboard after Frame 1

The symbol for a spare is ╱

1	
3	╱

The total score after the spare cannot be worked out until Mike bowls his next ball.

A spare scores 10 + the number of pins knocked over with the player's next ball

Here is the scoreboard after Frame 2

1		2	
3	╱	6	2
16		24	
↑ 10 + 6		↑ 16 + 6 + 2	

After Frame 1, Mike's total score is 16

After Frame 2, Mike's total score is 24

Example 3 – Fatima's first two frames

Frame 1

1st ball → 10 pins
strike

Frame 2

1st ball → 5 pins 2nd ball → 4 pins

Here is the scoreboard after Frame 1

The symbol for a strike is **X**

1
X

The total score after the strike cannot be worked out until Fatima bowls her next **two** balls.

A strike scores 10 + the number of pins knocked over with the player's next **two** balls

Here is the scoreboard after Frame 2

1		2	
X		5	4
19		28	

↑ 10 + 5 + 4 ↑ 19 + 5 + 4

After Frame 1, Fatima's total score is 19
After Frame 2, Fatima's total score is 28

The scoreboard

Here is the scoreboard for a complete game between Sophie, Mike and Fatima.

	1		2		3		4		5		6		7		8		9		10	
Sophie	6	2	5	4	3	/	3	4	**X**		6	3	4	/	7	1	1	6	3	4
	8		17		30		37		56		65		82		90		97		**104**	
Mike	3	/	6	2	2	/	5	4	**X**		6	/	4	3	7	2	**X**		6	3
	16		24		39		48		68		82		89		98		117		**126**	
Fatima	**X**		5	4	4	/	6	2	**X**		2	/	3	3	7	2	4	/	8	1
	19		28		44		52		72		85		91		100		118		**127**	

Sophie's final score is 104
Mike's final score is 126
Fatima's final score is 127

Fatima wins the game.

Data Sheet for Electricity

Electrical items use different amounts of electricity.

The electricity used depends on

the power of the item

the amount of time the item is switched on.

Power is measured in kilowatts (kW) or watts (W).

1 kW = 1000 W

Using electricity

The number of units of electricity an item uses, E, is worked out using the formula

$$E = PT$$

P is the power in kW

T is the time in hours

Example

A 3000 W electric fire is switched on for $2\frac{1}{2}$ hours.

P = 3000 ÷ 1000

= 3

T = 2.5

E = 3 × 2.5

= 7.5

The fire uses 7.5 units of electricity.

Cost of electricity used

Work out Number of units of electricity used × Cost per unit

END OF DATA

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Answer **all** questions in the spaces provided.

1 **Tenpin bowling**

There is a **data sheet** for Tenpin bowling.



I am captain of a tenpin bowling team.

QI Worked Solution:
See official AQA mark scheme method.

Anna

Anna is playing a game against Ben.

†1 (a) Anna has completed two frames of the game.

[Method & Accuracy]

1		2	
7	/	8	1

Work out her total score after Frame 2

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Do not write outside the box

1 (b) Later, Ben has one more frame to play.

1		2		3		4		5		6		7		8		9		10	
4	3	8	/	4	5	**X**		7	2	1	7	5	4	**X**		6	/		
7		21		30		49		58		66		75							

In Frame 10, Ben knocks over

5 pins with his first ball

3 pins with his second ball.

He says,

"My final score is **more than** 115"

Is he correct?
You **must** show your working.

[4 marks]

Question 1 continues on the next page

Do not write outside the box

Anna has to choose a new player for her bowling team.

She can choose either Jamil or Tom.

Jamil and Tom have each played some games against Anna.

Here are their final scores and results.

Jamil	
Final score	**Result**
155	Lost
147	Lost
216	Won
182	Lost
179	Won
177	Lost

Tom	
Final score	**Result**
191	Won
160	Lost
134	Lost
210	Won
182	Lost
202	Won
159	Lost
146	Lost

Do not write outside the box

1 (c)



I have a greater chance of winning against Anna than Jamil does.

Tom

Is Tom correct?
You **must** show your working.

[3 marks]

1 (d)



On average, my final scores were higher than Tom's.

Jamil

Is Jamil correct?
You **must** show your working.

[4 marks]

14

2 Sandwiches



I make and sell sandwiches in my shop.

Amir

2 (a) Amir uses 300 grams of spread to make 25 sandwiches.

He buys the spread in 2.5 kg tubs.

He has no spread left.

Next week, Amir wants to make 500 sandwiches.

How many tubs does he need to buy?
You **must** show your working.

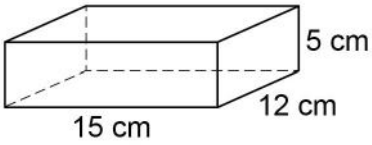
[5 marks]

Q2 Worked Solution:

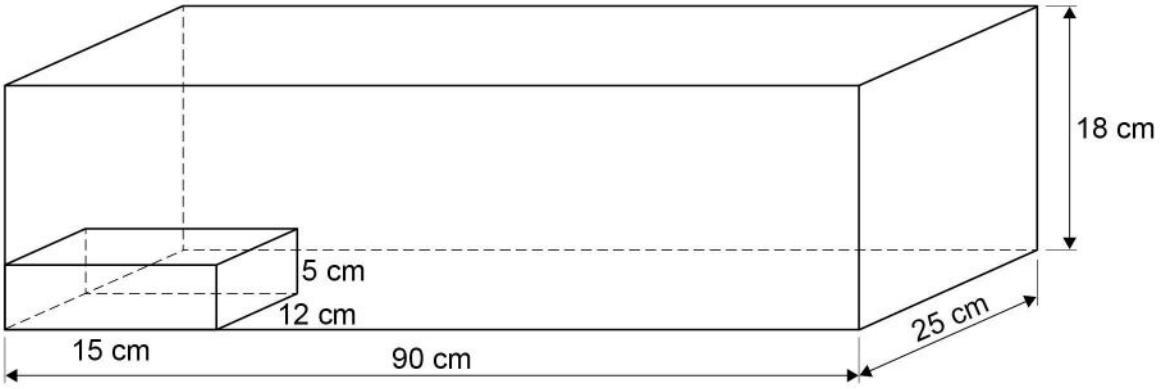
182
their
160
their
□
or 171
must have attempted to order the data
178 and 171 and Yes
1b 1b
A1 178 and 171
[M1] [Rc] [A2]

Do not write outside the box

2 (b) Amir puts the sandwiches in cuboid boxes.



The boxes of sandwiches are packed in cuboid crates.
The boxes are all packed in the same way, as shown.



Work out the **maximum** number of boxes that can be packed in one crate.
You **must** show your working.

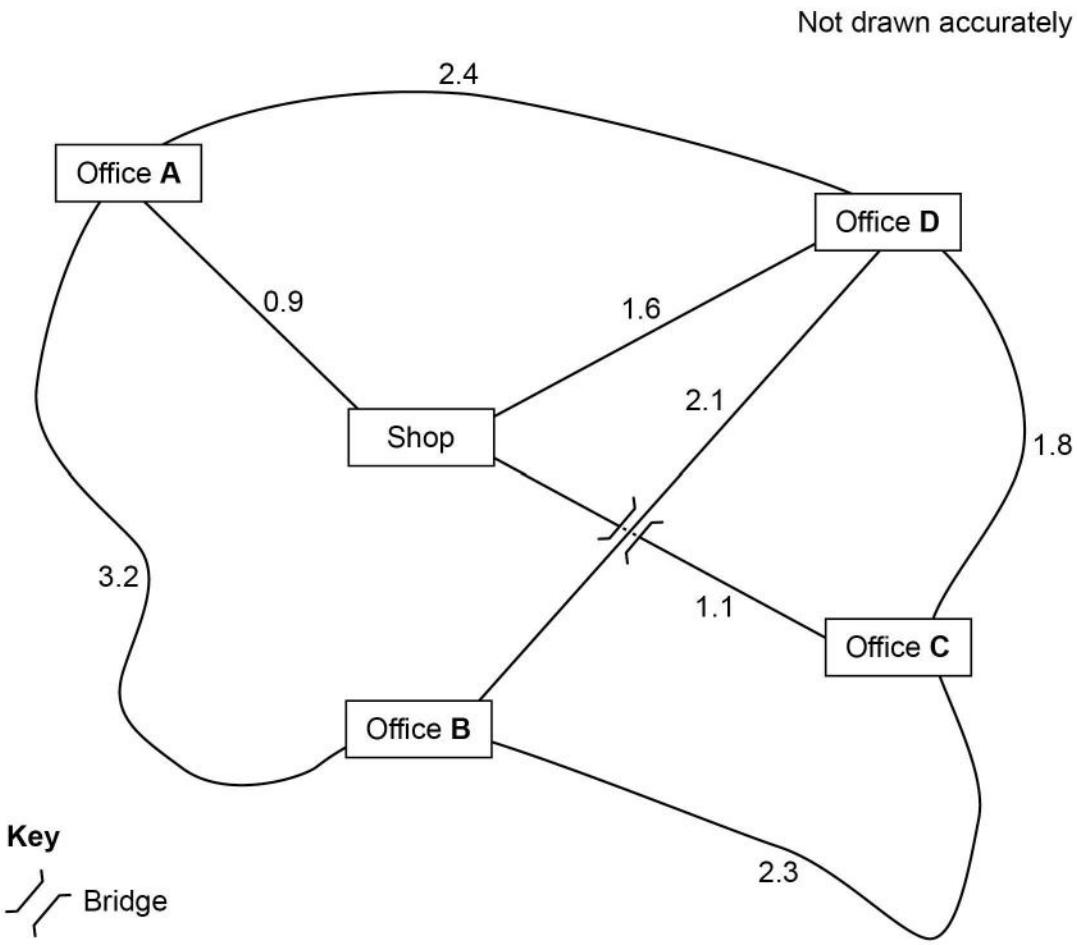
[4 marks]

Do not write outside the box

Amir delivers sandwiches to four offices.

The diagram shows the driving distances between his shop and the offices.

The distances are in miles.



The road between the shop and Office C goes under the bridge.

Do not write outside the box

2 (c) On Monday, Amir drives

Shop ⟶ Office D ⟶ Office C ⟶ Shop

How many miles does he drive?
Circle your answer.

[1 mark]

3.4 4.3 4.5 5.4

2 (d) On Tuesday, Amir drives from his shop.

He visits each of the four offices.

He then drives back to the shop.

Work out the shortest possible route.
Include the route and the total distance he drives.

[3 marks]

13

3 Hairdressing salon



I am opening a hairdressing salon.

Jenny

3 (a) Jenny is designing the salon.

Each sink will need a square space with sides of 50 cm

Each chair will need a circular space with radius 50 cm

She wants

- three sinks against the same wall, at least 1 m apart
- four chairs in a line, each 1 m from one wall
- a rectangular reception desk measuring 1 m by 50 cm
- a rectangular display cabinet measuring 2 m by 1 m
- a rectangular waiting area measuring 2 m by 1.5 m
- the door to be able to open fully.

Show a possible design on the scale drawing opposite.

[6 marks]

Q3 Worked Solution:

Marks are awarded for demonstrating the following interrelated process skills.
Representing
Selecting the mathematics and information to model a situation.
R.1
Candidates recognise that a situation has aspects that can be represented using mathematics.
R.2
Candidates make an initial model of a situation using suitable forms of representation.
R.3
[Method & Accuracy]

Do not write outside the box

Here are the people who work in the salon.

Stylists			
Jenny	Craig	Fay	Mia

Juniors	
Erik	Wendy

3 (b) The salon opens six days each week, from Monday to Saturday.

There are two shifts each day.

From Monday to Friday, two stylists and one junior work each shift.

On Saturday, three stylists and one junior work each shift.

Each week

- Jenny, Mia and Erik each work 8 shifts
- Craig and Fay each work 5 shifts
- Craig does **not** work on Saturday
- Wendy works 4 shifts, all in the afternoon
- each **stylist** has at least 1 full day off.

Make a possible rota for one week.
Use the grids opposite.

[4 marks]

Practise on this grid.

	Morning shift				Afternoon shift			
	Stylist 1	**Stylist 2**	**Stylist 3**	**Junior**	**Stylist 1**	**Stylist 2**	**Stylist 3**	**Junior**
Monday								
Tuesday								
Wednesday								
Thursday								
Friday								
Saturday								

Put your answer on this grid.

	Morning shift				Afternoon shift			
	Stylist 1	**Stylist 2**	**Stylist 3**	**Junior**	**Stylist 1**	**Stylist 2**	**Stylist 3**	**Junior**
Monday								
Tuesday								
Wednesday								
Thursday								
Friday								
Saturday								

3 (c) Here are the prices at the salon.

Cut and blow dry	£33
Cut and colour	£60

In the first week the salon has these appointments

35 for a cut and blow dry
15 for a cut and colour.

The table shows the number of appointments for Craig, Fay and Mia.

	Cut and blow dry	**Cut and colour**
Craig	5	2
Fay	4	3
Mia	16	4

The rest of the appointments are for Jenny.

In the first week Jenny will pay

- 10% of the price of their own appointments to Craig, Fay and Mia
- £980 in other wages
- £325 in other costs.

Work out Jenny's profit in the first week.

[6 marks]

Do not write outside the box

4 **Electricity**

There is a **data sheet** for Electricity.

4 (a) Joe has an electric fire with three settings.

Setting	High	Medium	Low
Power	2 kW	1.5 kW	1 kW

Q4 Worked Solution:

1 × 13.88 – 4

Last week, he used the fire on

1 × their 7.12

High for 6 hours on 1 night
Medium for 5 hours on each of 4 nights
Low for 3 hours on each of 2 nights.

Electricity costs 15.5 pence per unit.

Joe says,

"Last week, the cost of electricity for using the fire was less than £10"

Is he correct?
You **must** show your working.

[6 marks]

or 3.47 – 1.78 or 1.69
their 7.12 from (b)
~~difference in cost of bulbs~~
2.97(6) and 1.69 and Yes
A2ft
lb lb
~~ft their 7.12 from (b)~~
A1ft 2.97(6) and 1.69
[M1] [A0]

Do not write outside the box

Joe has a lamp that uses one light bulb.

†4 (b) An ordinary 100 W bulb

- lasts for 6 months
- costs 89 pence.

How much will Joe pay to buy ordinary 100 W bulbs for his lamp for **4 years**?

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Question 4 continues on the next page

Do not write outside the box

4 (c) Joe looks at the cost of buying and using ordinary 100 W bulbs or a low energy bulb for the next **4 years**.

He makes these notes.

The lamp is switched on for 240 hours **each year**.

Electricity costs 15.5 pence per unit.

A low energy bulb

- lasts for 4 years
- costs £13.88
- uses 20% of the number of units of electricity used by an ordinary 100 W bulb.

He says,

"The total cost is less if I use a low energy bulb."

Is he correct?
You **must** show your working.

[8 marks]

END OF QUESTIONS

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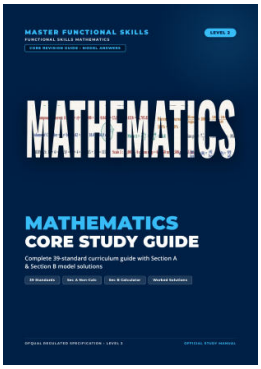
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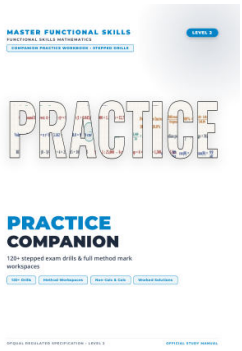


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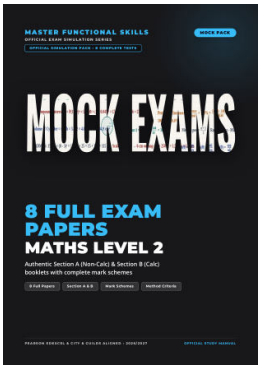


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